

Problem 5 (Caucasus MO 2021 Juniors Problem 3). There are $n > 2$ nonzero integers such that each of them is divisible by the sum of the other $n - 1$ numbers. Prove that the sum of all these integers is 0.

Solution. Let the integers be $a_1, a_2, \dots, a_n$, and let $S = a_1 + a_2 + \dots + a_n$. For each $k = 1, 2, \dots, n$, we have $S - a_k \mid a_k$ by the given condition. Since $a_k \neq 0$, we have $|S - a_k| \leq |a_k|$. Squaring both sides, we get $S^2 - 2a_k S + a_k^2 \leq a_k^2$, and hence $S^2 \leq 2a_k S$. Summing over all k 's, we obtain $nS^2 \leq 2(a_1 + a_2 + \dots + a_n)S = 2S^2$. As $n > 2$, we must have $S = 0$. $\square$

Problem 6 (Indian TST 1998). Let S be a set of $2k$ elements where k is a positive integer. Let $\mathcal{F}$ be a family of subsets of S such that every set in $\mathcal{F}$ has exactly k elements. Suppose every subset of S having exactly $k - 1$ elements is a subset of exactly one element in $\mathcal{F}$. Prove that $k + 1$ is a prime number.

Solution. WLOG assume $S = \{1, 2, \dots, 2k\}$. Suppose on the contrary that $k + 1$ is not a prime. Since $k + 1 > 1$, it must have a prime divisor p . Clearly, $p < k$. We count the number of $(k - 1)$ -element subsets of S containing $1, 2, \dots, k - p$ (which we call *good subsets*).

On the one hand, since we need to choose $(k - 1) - (k - p) = p - 1$ more elements from the remaining $2k - (k - p) = k + p$ elements, there are $\binom{k+p}{p-1}$ good subsets.

On the other hand, suppose there are x elements in $\mathcal{F}$ which contains all of $1, 2, \dots, k - p$. Note that all good subsets are subsets of exactly one of these x elements. Also, each of these x elements contains exactly $\binom{p}{p-1} = p$ good subsets. Thus, the total number of good subsets is px .

It follows that $px = \frac{(k+p)(k+p-1) \cdots (k+2)}{(p-1)!}$, and hence $p \mid (k+p)(k+p-1) \cdots (k+2)$. But this is impossible since $k + p - m = (k + 1) + p - (m + 1) \equiv -(m + 1) \not\equiv 0 \pmod{p}$ for $m = 0, 1, \dots, p - 2$. This is a contradiction. So $k + 1$ is a prime. $\square$

Problem 7 (239 Open MO 2014 Problem 6). Let $a_1, a_2, \dots, a_n$ be positive real numbers satisfying $a_1^2 + 2a_2^3 + \dots + na_n^{n+1} < 1$. Prove that

$$2a_1 + 3a_2^2 + \dots + (n+1)a_n^n < 3.$$

Solution. We claim that

$$\sum_{k=1}^n (k+1)a_k^k < 2 \sum_{k=1}^n ka_k^{k+1} + 1$$

for any positive real numbers $a_1, a_2, \dots, a_n$. This obviously yields the desired inequality.

Indeed, for each k , by the AM-GM inequality, we have

$$2ka_k^{k+1} + \frac{1}{2^k} = \underbrace{2a_k^{k+1} + 2a_k^{k+1} + \dots + 2a_k^{k+1}}_{k \text{ times}} + \frac{1}{2^k} \geq (k+1) \sqrt[k+1]{(2a_k^{k+1})^k \left(\frac{1}{2^k}\right)} = (k+1)a_k^k.$$

Summing over $k = 1, 2, \dots, n$, we obtain

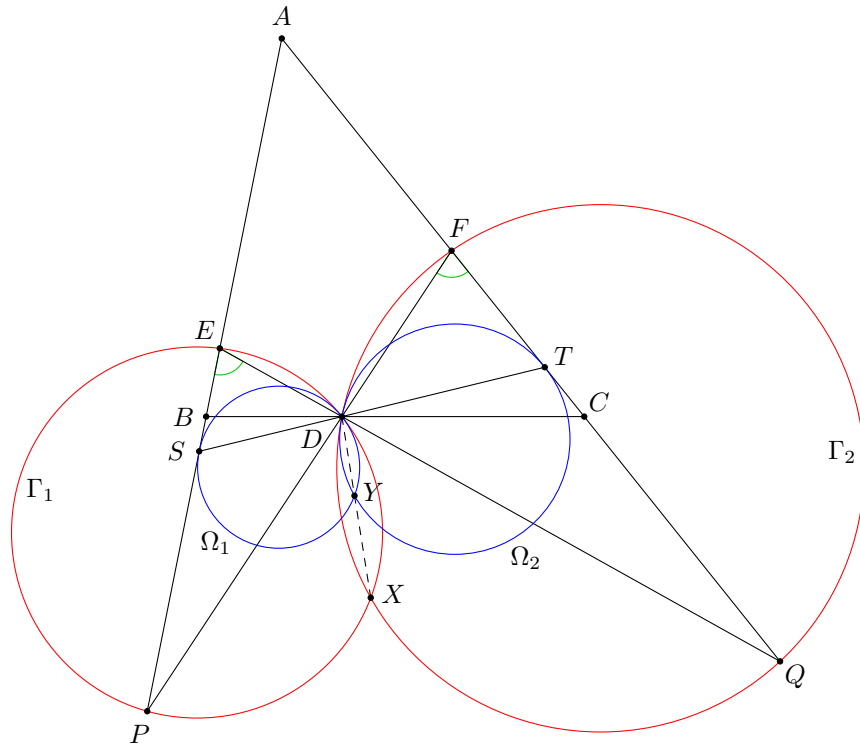
$$\sum_{k=1}^n (k+1)a_k^k \leq 2 \sum_{k=1}^n ka_k^{k+1} + \sum_{k=1}^n \frac{1}{2^k} < 2 \sum_{k=1}^n ka_k^{k+1} + \sum_{k=1}^{\infty} \frac{1}{2^k} = 2 \sum_{k=1}^n ka_k^{k+1} + 1$$

as desired. $\square$

Problem 8 (Vietnam MO 2018 Day 1 Problem 2). Let $\triangle ABC$ be a scalene acute triangle. Let D be a point on the side BC . Points E and F are chosen on the sides AB and AC respectively such that $\angle DEB = \angle DFC$. Define the following.

- P is the intersection point of the extensions of FD and AB
- Q is the intersection point of the extensions of ED and AC
- Γ_1 is the circumcircle of $\triangle DEP$
- Γ_2 is the circumcircle of $\triangle DFQ$
- Ω_1 is the circle internally tangent to Γ_1 at D and tangent to the line AB at S
- Ω_2 is the circle internally tangent to Γ_2 at D and tangent to the line AC at T
- X is the intersection point of Γ_1 and Γ_2 different from D
- Y is the intersection point of Ω_1 and Ω_2 different from D
- K is the intersection point of the circumcircles of $\triangle AEF$ and $\triangle AST$ different from A
- L is the intersection point of the circumcircle of $\triangle AEF$ and the line AY different from A

- (a) Prove that D, X, Y are collinear.
- (b) Prove that the tangent at D to the circumcircle of $\triangle DYK$ meets EF at a point on the circumcircle of $\triangle DKL$.



Solution. We only provide a proof for the configuration as shown. Other cases are similar.

- (a) Since $\angle PED = \angle QFD$ and $\angle PDE = \angle QDF$, we have $\triangle PDE \sim \triangle QDF$. Since Ω_1 is tangent to $\Gamma_1 = (PDE)$ at D and tangent to PE , while Ω_2 is tangent to $\Gamma_2 = (QDF)$ at D and tangent to QF , the circles Ω_1 and Ω_2 must be corresponding circles under the similarity of the two triangles. Thus, the homothety with centre D mapping Ω_1 to Γ_1 is the same as the homothety with centre D mapping Ω_2 to Γ_2 . This homothety must map Y to X . So D, Y, X are collinear.
- (b) Let the tangent at D to (DYK) meet EF at W . Firstly, since $\angle PEQ = \angle PFQ$, the points P, Q, F, E are concyclic. Therefore, $PE \cap QF = A$ lies on the radical axis DX of Γ_1 and Γ_2 . This implies A, D, Y, X are collinear.

Secondly, it is well-known that DS is the angle bisector of $\angle PDE$. (Consider the homothety mapping Ω_1 to Γ_1 . Then S is mapped to the midpoint of the arc PE since the tangent at S to Ω_1 is parallel to the tangent at the image to Γ_1 .) Similarly, DT is the angle bisector of $\angle QDF$. Thus, S, D, T are collinear.

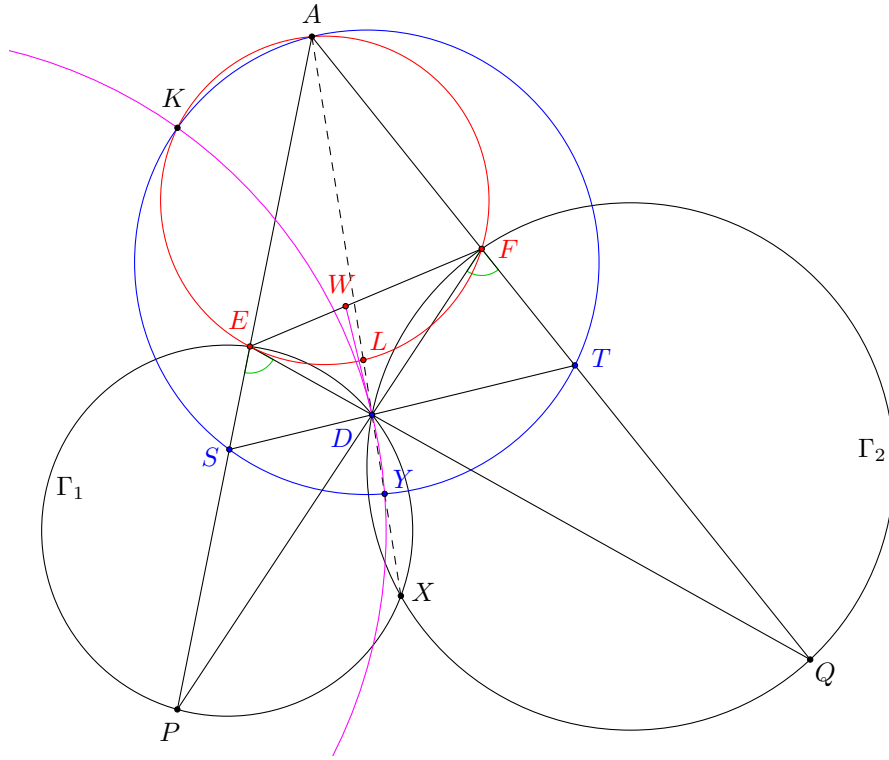
Now, since the intersection points of $(AELF)$ and $(ASYT)$ are A and K , and AES, ALY, AFT are collinear, we see that K is the centre of spiral similarity mapping $\triangle ELF$ to $\triangle SYT$. Thus, K is also the centre of spiral similarity mapping ES to FT . Moreover, since W and D lie on EF and ST respectively, and

$$\angle KDW = \angle KYD = \angle KYL = \angle KSE,$$

the spiral similarity mapping EF to ST also maps W to D . Thus, we have $\triangle KWL \sim \triangle KDY$. It follows that

$$\angle K LW = \angle KYD = \angle KDW,$$

so that W lies on (DKL) as desired.



□